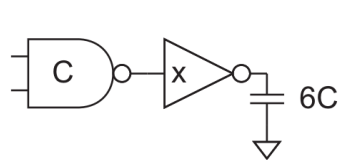


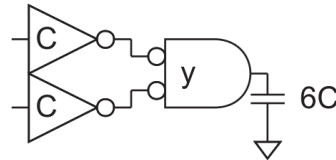
Tutorial 7

For all questions in this tutorial, unless specified, you can assume that the smallest inverter in the technology is sized 2/1 for matched rising and falling delays. Unless otherwise mentioned assume that all shared S/D do not have a cap reduction.

1. Compute the rising and falling propagation delays of the NOR gate driving h identical NOR gates using the Elmore delay model. Assume that every source or drain has fully contacted diffusion when making your estimate of capacitance
2. Sketch a stick diagram for the 2-input NOR. Compute the rising and falling propagation delays of the NOR gate driving h identical NOR gates using the Elmore delay model with better capacitance estimates. In particular, if a diffusion node is shared between two parallel transistors, only budget its capacitance once. If a diffusion node is between two series transistors and requires no contacts, only budget half the capacitance because of the smaller diffusion area.
3. Find the worst-case Elmore parasitic delay of an n -input NOR gate
4. Consider the two designs for a 2-input AND gate. Give an intuitive argument as to which will be faster. Back up your argument with a calculation of the path effort, delay and the input caps x and y needed to achieve this delay

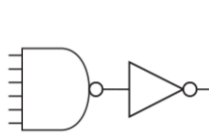


(a)

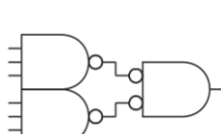


(b)

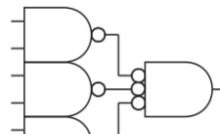
5. Consider four designs of a 6-input AND gate shown in Figure below. Develop an expression for the delay of each path if the path electrical effort is H . What design is fastest for $H=1$? For $H=5$? For $H=20$? Explain your conclusions intuitively.



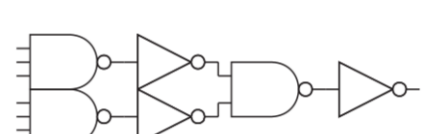
(a)



(b)



(c)

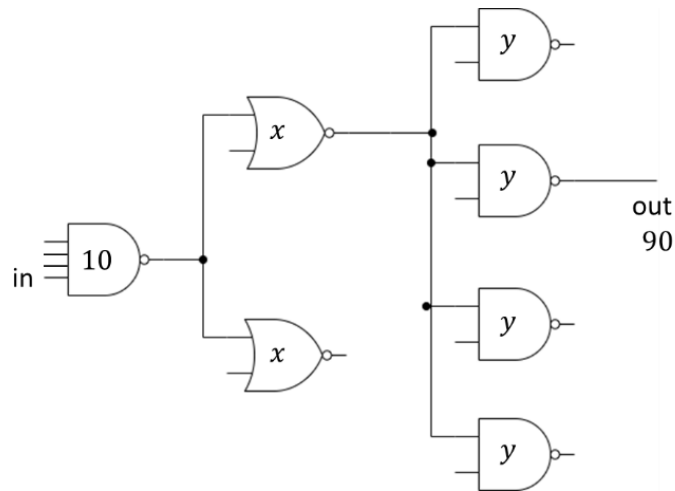


(d)

6. Design a circuit at a gate level to compute the following function

```
if (a==b) y=a;
else y=0;
```

Let a , b and y be 16-bit busses. Assume that the input and output capacitance are each 10 units. Your goal is to make the circuit as fast as possible. Estimate the delay in FO4 inverter delays. What sizes do you need to achieve this delay?
7. Size the path in the figure for minimum delay



8. A 3-stage logic path is designed so that the effort borne by each stage is 12, 6, and 9 delay units, respectively. Can this design be improved? Why? What is the best number of stages for this path? What changes do you recommend to the existing design?
9. An output pad contains a chain of successively larger inverters to drive the large off-chip capacitance. If the first inverter in the chain has an input capacitance of 20fF and the off-chip load is 10pF, how many inverters should be used to drive the load with the least delay? Estimate the delay, expressed in FO4 inverter delays.